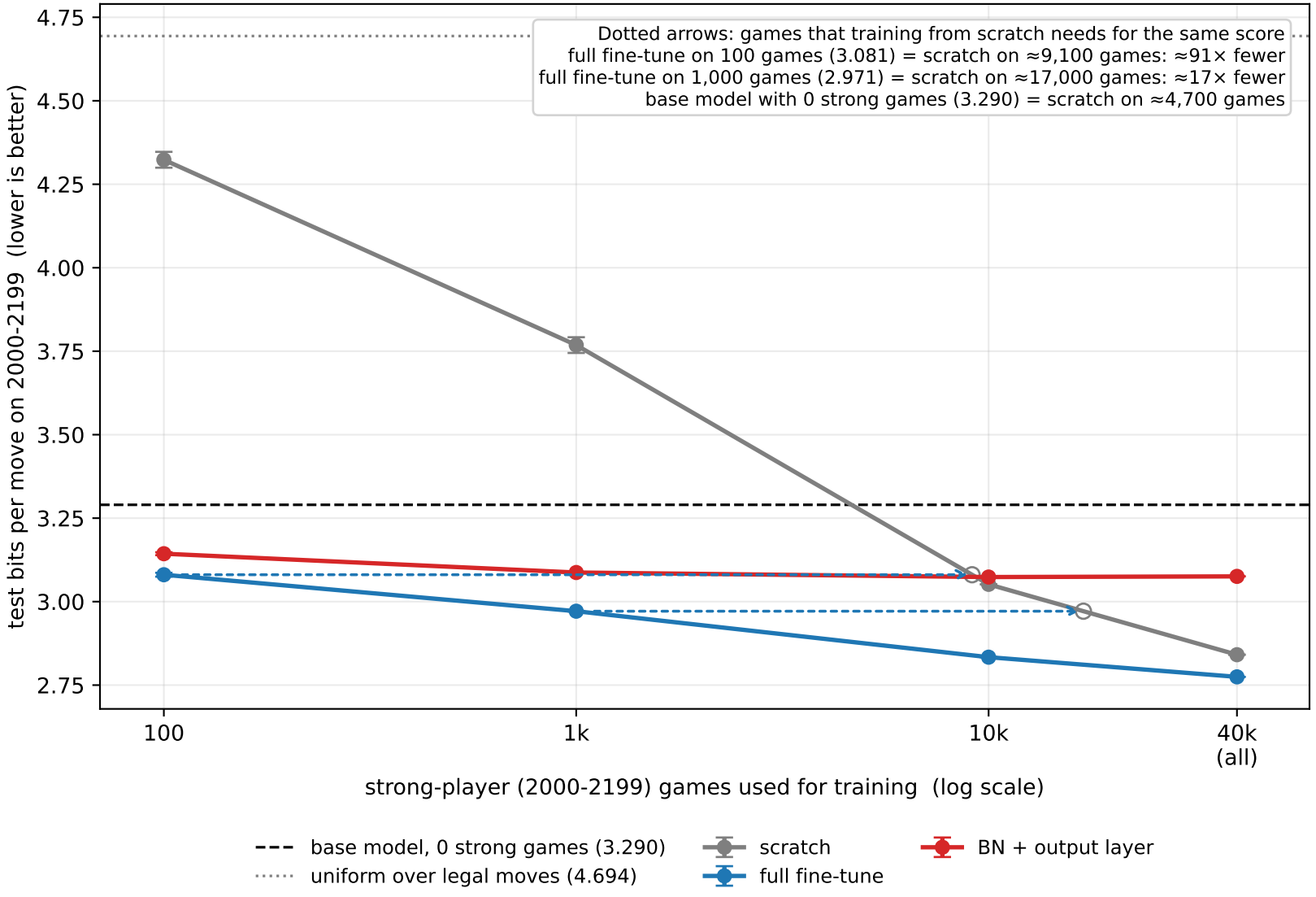
Fine-tuning needs ≈17-91× fewer strong games than untuned scratch for the same score



Test bits per move on 37,050 held-out 2000-2199 positions (lower is better), against how many 2000-2199 games each model was trained on. Scratch = random start; fine-tunes start from the base model trained on 141,333 games of 1000-1199 players. Scratch was not tuned for small N (lr 1e-3, no regularisation), and every option picks its checkpoint on 42,433 val positions from 540 strong games, more games than the N = 100 training set. Error bars: ±1 standard deviation over 3 seeds (N = 100 and 1k); one seed at 10k and 40k. Savings are interpolated linearly in log(N) between the measured scratch points, so they are rough.